🔺 Mersenne Deterministic Generator — True Grammar

$p\_e = m^2 \times (p\_a \times p\_b) + k$ | $k = p\_c - p\_d$ (pure, no multiplier) | Negative k OK | Unique products

51

Found/52

136,279,841

Searching

94

m Used

2p+p

k-Type

⚠️ NOT FOUND: M42=25,964,951, M52=136,279,841

Ready. Seeds: M1=2, M2=3

🔑 TRUE GRAMMAR: k = p_c − p_d (pure, no multiplier) | Negative k allowed | Unique products

🌱 Seeds: M1=2, M2=3

🔑 TRUE GRAMMAR: k = p_c−p_d (pure, no multiplier)

✅ Negative k allowed | Unique (p_a,p_b) pairs | 4 k-types

═══════ SEARCHING M3 = 5 ═══════

Using 2 primes | Unique products: 0

✅ FOUND M3!

$= 1^2 \times (2 \times 2) - (M1-M2)$

k = -1 [diff]

Total: 3 primes | Unique products: 1

═══════ SEARCHING M4 = 7 ═══════

Using 3 primes | Unique products: 1

✅ FOUND M4!

$= 1^2 \times (2 \times 3) - (M1-M2)$

k = -1 [diff]

Total: 4 primes | Unique products: 2

═══════ SEARCHING M5 = 13 ═══════

Using 4 primes | Unique products: 2

✅ FOUND M5!

$= 1^2 \times (2 \times 5) - (M1-M3)$

k = -3 [diff]

Total: 5 primes | Unique products: 3

══════ SEARCHING M6 = 17 ══════

Using 5 primes | Unique products: 3

✅ FOUND M6!

$= 1^2 \times (2 \times 7) - (M1 - M3)$

k = -3 [diff]

Total: 6 primes | Unique products: 4

══════ SEARCHING M7 = 19 ══════

Using 6 primes | Unique products: 4

✅ FOUND M7!

$= 1^2 \times (2 \times 13) - (2 \times M1 + M2)$

k = 7 [2p+p]

Total: 7 primes | Unique products: 5

══════ SEARCHING M8 = 31 ══════

Using 7 primes | Unique products: 5

✅ FOUND M8!

$= 1^2 \times (2 \times 17) + (M1 - M3)$

k = -3 [diff]

Total: 8 primes | Unique products: 6

══════ SEARCHING M9 = 61 ══════

Using 8 primes | Unique products: 6

✅ FOUND M9!

$= 1^2 \times (2 \times 19) + (2 \times M1 + M7)$

k = 23 [2p+p]

Total: 9 primes | Unique products: 7

══════ SEARCHING M10 = 89 ══════

Using 9 primes | Unique products: 7

✅ FOUND M10!

$= 1^2 \times (2 \times 31) - (2 \times M1 - M8)$

k = -27 [2p-p]

Total: 10 primes | Unique products: 8

══════ SEARCHING M11 = 107 ══════

Using 10 primes | Unique products: 8

✅ FOUND M11!

$= 1^2 \times (2 \times 61) + (M1 - M6)$

k = -15 [diff]

Total: 11 primes | Unique products: 9

════════ SEARCHING M12 = 127 ════════

Using 11 primes | Unique products: 9

✅ FOUND M12!

$= 1^2 \times (2 \times 89) + (2 \times M3 - M9)$

k = -51 [2p-p]

Total: 12 primes | Unique products: 10

════════ SEARCHING M13 = 521 ════════

Using 12 primes | Unique products: 10

✅ FOUND M13!

$= 1^2 \times (2 \times 127) + (2 \times M12 + M5)$

k = 267 [2p+p]

Total: 13 primes | Unique products: 11

════════ SEARCHING M14 = 607 ════════

Using 13 primes | Unique products: 11

✅ FOUND M14!

$= 2^2 \times (2 \times 107) - (2 \times M9 + M12)$

k = 249 [2p+p]

Total: 14 primes | Unique products: 12

════════ SEARCHING M15 = 1,279 ════════

Using 14 primes | Unique products: 12

✅ FOUND M15!

$= 1^2 \times (2 \times 521) + (2 \times M12 - M6)$

k = 237 [2p-p]

Total: 15 primes | Unique products: 13

════════ SEARCHING M16 = 2,203 ════════

Using 15 primes | Unique products: 13

✅ FOUND M16!

$= 10^2 \times (3 \times 3) + (2 \times M14 + M10)$

k = 1,303 [2p+p]

Total: 16 primes | Unique products: 14

════════ SEARCHING M17 = 2,281 ════════

Using 16 primes | Unique products: 14

✅ FOUND M17!

$= 2^2 \times (2 \times 607) - (2 \times M15 + M6)$

k = 2,575 [2p+p]

Total: 17 primes | Unique products: 15

═══════ SEARCHING M18 = 3,217 ═══════

Using 17 primes | Unique products: 15

✅ FOUND M18!

$= 14^2 \times (3 \times 5) + (2 \times M15 - M17)$

k = 277 [2p-p]

Total: 18 primes | Unique products: 16

═══════ SEARCHING M19 = 4,253 ═══════

Using 18 primes | Unique products: 16

✅ FOUND M19!

$= 1^2 \times (2 \times 2203) - (2 \times M5 + M12)$

k = 153 [2p+p]

Total: 19 primes | Unique products: 17

═══════ SEARCHING M20 = 4,423 ═══════

Using 19 primes | Unique products: 17

✅ FOUND M20!

$= 1^2 \times (2 \times 2281) - (2 \times M9 + M6)$

k = 139 [2p+p]

Total: 20 primes | Unique products: 18

═══════ SEARCHING M21 = 9,689 ═══════

Using 20 primes | Unique products: 18

✅ FOUND M21!

$= 1^2 \times (2 \times 3217) + (2 \times M7 + M18)$

k = 3,255 [2p+p]

Total: 21 primes | Unique products: 19

═══════ SEARCHING M22 = 9,941 ═══════

Using 21 primes | Unique products: 19

✅ FOUND M22!

$= 16^2 \times (3 \times 7) + (2 \times M17 + M2)$

k = 4,565 [2p+p]

Total: 22 primes | Unique products: 20

═══════ SEARCHING M23 = 11,213 ═══════

Using 22 primes | Unique products: 20

✅ FOUND M23!

$= 2^2 \times (2 \times 1279) + (2 \times M13 - M9)$

k = 981 [2p-p]

Total: 23 primes | Unique products: 21

═══════ SEARCHING M24 = 19,937 ═══════

Using 23 primes | Unique products: 21

✅ FOUND M24!

$= 1^2 \times (2 \times 4423) - (2 \times M9 - M23)$

k = -11,091 [2p-p]

Total: 24 primes | Unique products: 22

═══════ SEARCHING M25 = 21,701 ═══════

Using 24 primes | Unique products: 22

✅ FOUND M25!

$= 1^2 \times (2 \times 19937) - (2 \times M23 - M19)$

k = 18,173 [2p-p]

Total: 25 primes | Unique products: 23

═══════ SEARCHING M26 = 23,209 ═══════

Using 25 primes | Unique products: 23

✅ FOUND M26!

$= 26^2 \times (3 \times 13) + (2 \times M8 - M18)$

k = -3,155 [2p-p]

Total: 26 primes | Unique products: 24

═══════ SEARCHING M27 = 44,497 ═══════

Using 26 primes | Unique products: 24

✅ FOUND M27!

$= 1^2 \times (2 \times 21701) - (2 \times M20 - M22)$

k = -1,095 [2p-p]

Total: 27 primes | Unique products: 25

═══════ SEARCHING M28 = 86,243 ═══════

Using 27 primes | Unique products: 25

✅ FOUND M28!

$= 3^2 \times (2 \times 4253) + M21$

k = 9,689 [single]

Total: 28 primes | Unique products: 26

═══════ SEARCHING M29 = 110,503 ═══════

Using 28 primes | Unique products: 26

✅ FOUND M29!

$= 42^2 \times (3 \times 19) + (2 \times M4 + M22)$

k = 9,955 [2p+p]

Total: 29 primes | Unique products: 27

═══════ SEARCHING M30 = 132,049 ═══════

Using 29 primes | Unique products: 27

✅ FOUND M30!

= $50^2 \times (3 \times 17) + (2 \times M17 - M5)$

k = 4,549 [2p-p]

Total: 30 primes | Unique products: 28

═══════ SEARCHING M31 = 216,091 ═══════

Using 30 primes | Unique products: 28

✅ FOUND M31!

= $48^2 \times (3 \times 31) - (2 \times M22 - M25)$

k = -1,819 [2p-p]

Total: 31 primes | Unique products: 29

═══════ SEARCHING M32 = 756,839 ═══════

Using 31 primes | Unique products: 29

✅ FOUND M32!

= $174^2 \times (5 \times 5) - M9$

k = 61 [single]

Total: 32 primes | Unique products: 30

═══════ SEARCHING M33 = 859,433 ═══════

Using 32 primes | Unique products: 30

✅ FOUND M33!

= $82^2 \times (5 \times 19) + (2 \times M17 + M31)$

k = 220,653 [2p+p]

Total: 33 primes | Unique products: 31

═══════ SEARCHING M34 = 1,257,787 ═══════

Using 33 primes | Unique products: 31

✅ FOUND M34!

= $84^2 \times (3 \times 61) - (2 \times M25 - M22)$

k = 33,461 [2p-p]

Total: 34 primes | Unique products: 32

═══════ SEARCHING M35 = 1,398,269 ═══════

Using 34 primes | Unique products: 32

✅ FOUND M35!

= $66^2 \times (3 \times 107) - (2 \times M1 + M2)$

k = 7 [2p+p]

Total: 35 primes | Unique products: 33

═══════ SEARCHING M36 = 2,976,221 ═══════

Using 35 primes | Unique products: 33

✅ FOUND M36!

= $128^2 \times (3 \times 89) - (2 \times M7 + M35)$

k = 1,398,307 [2p+p]

Total: 36 primes | Unique products: 34

═══════ SEARCHING M37 = 3,021,377 ═══════

Using 36 primes | Unique products: 34

✅ FOUND M37!

= $118^2 \times (7 \times 31) - (2 \times M1 + M12)$

k = 131 [2p+p]

Total: 37 primes | Unique products: 35

═══════ SEARCHING M38 = 6,972,593 ═══════

Using 37 primes | Unique products: 35

✅ FOUND M38!

= $286^2 \times (5 \times 17) - (2 \times M1 - M24)$

k = -19,933 [2p-p]

Total: 38 primes | Unique products: 36

═══════ SEARCHING M39 = 13,466,917 ═══════

Using 38 primes | Unique products: 36

✅ FOUND M39!

= $148^2 \times (5 \times 127) - (2 \times M31 + M22)$

k = 442,123 [2p+p]

Total: 39 primes | Unique products: 37

═══════ SEARCHING M40 = 20,996,011 ═══════

Using 39 primes | Unique products: 37

✅ FOUND M40!

= $420^2 \times (7 \times 17) + (2 \times M16 + M3)$

k = 4,411 [2p+p]

Total: 40 primes | Unique products: 38

═══════ SEARCHING M41 = 24,036,583 ═══════

Using 40 primes | Unique products: 38

✅ FOUND M41!

= $233^2 \times (7 \times 61) - (M19 - M33)$

k = -855,180 [diff]

Total: 41 primes | Unique products: 39

══════ SEARCHING M42 = 25,964,951 ══════

Using 41 primes | Unique products: 39

❌ NOT FOUND with m 5000

🔼 Increasing m: 5000 → 10000, retrying...

⚠️ Marked as MISSING.

══════ SEARCHING M43 = 30,402,457 ══════

Using 41 primes | Unique products: 39

✅ FOUND M43!

$= 35^2 \times (2 \times 9941) + (2 \times M37 + M19)$

k = 6,047,007 [2p+p]

Total: 42 primes | Unique products: 40

══════ SEARCHING M44 = 32,582,657 ══════

Using 42 primes | Unique products: 40

✅ FOUND M44!

$= 41^2 \times (2 \times 9689) + (2 \times M20 - M14)$

k = 8,239 [2p-p]

Total: 43 primes | Unique products: 41

══════ SEARCHING M45 = 37,156,667 ══════

Using 43 primes | Unique products: 41

✅ FOUND M45!

$= 4^2 \times (107 \times 21701) + (2 \times M17 - M4)$

k = 4,555 [2p-p]

Total: 44 primes | Unique products: 42

══════ SEARCHING M46 = 42,643,801 ══════

Using 44 primes | Unique products: 42

✅ FOUND M46!

$= 219^2 \times (7 \times 127) - (M18 - M21)$

k = -6,472 [diff]

Total: 45 primes | Unique products: 43

══════ SEARCHING M47 = 43,112,609 ══════

Using 45 primes | Unique products: 43

✅ FOUND M47!

$= 10^2 \times (5 \times 7) + (2 \times M36 + M44)$

k = 43,109,109 [2p+p]

Total: 46 primes | Unique products: 44

══════ SEARCHING M48 = 57,885,161 ══════

Using 46 primes | Unique products: 44

✅ FOUND M48!

$= 278^2 \times (7 \times 107) - (2 \times M6 + M13)$

k = 555 [2p+p]

Total: 47 primes | Unique products: 45

══════ SEARCHING M49 = 74,207,281 ══════

Using 47 primes | Unique products: 45

✅ FOUND M49!

$= 945^2 \times (5 \times 13) - (M40 - M44)$

k = -16,160,656 [diff]

Total: 48 primes | Unique products: 46

══════ SEARCHING M50 = 77,232,917 ══════

Using 48 primes | Unique products: 46

✅ FOUND M50!

$= 50^2 \times (61 \times 107) + (2 \times M42 + M29)$

k = 60,915,417 [2p+p]

Total: 49 primes | Unique products: 47

══════ SEARCHING M51 = 82,589,933 ══════

Using 49 primes | Unique products: 47

✅ FOUND M51!

$= 76^2 \times (3 \times 4423) + (2 \times M36 - M19)$

k = 5,948,189 [2p-p]

Total: 50 primes | Unique products: 48

══════ SEARCHING M52 = 136,279,841 ══════

Using 50 primes | Unique products: 48

✅ FOUND M52!

$= 94^2 \times (7 \times 2203) + (2 \times M22 + M2)$

k = 19,885 [2p+p]

Total: 51 primes | Unique products: 49